

# THOMAS FOURNIER

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## Automotive Mechanical Engineering Student | Seeking 6-Months Internship

### EDUCATION

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#### ISAE-Supméca

Paris, France

Master of Science in Mechanical Engineering

2024 – 2027

- Relevant Coursework: Mechanical design, CAD, Fluid Dynamics, Thermodynamics, Vehicle Dynamics

#### Claude Fauriel High School

Saint-Etienne, France

Two years intensive course in mathematics and physics

2022 - 2024

#### Claude Fauriel High School

Saint-Etienne, France

French scientific A-levels with honors

2022

### EXPERIENCE

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#### Universitat Politècnica de València

València, Spain

Research and Development intern

September 2025 – January 2026

- Contributed to the development of real-time SI engine combustion models to reproduce standard cycles.
- Developed chemical simulations to predict the delay of auto-ignition of air/fuel mixtures.
- Post-processed experimental data from an engine test bench and obtained a mean relative error of 3%.

#### Renault

Saint-Etienne, France

Automotive mechanic intern

July 2025

- Supported routine servicing tasks (brake systems, basic diagnostics).
- Contributed to complex repairs: engine replacement and full wiring harness installation.
- Developed teamwork, adherence to safety procedures and gained hands-on exposure to the automotive industry.

### PROJECTS

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#### Formula Student

September 2026 - Present

- Designed and tested the car's aerodynamic components (wings/aerofoils), tweaking geometry and positioning through multiple iterations to get the best downforce-to-drag balance to the vehicle.
- Modeled parts in 3DEXperience and ran CFD simulations in Ansys Fluent to check aero performance held up.
- Worked closely with other teams, demonstrating strong communication and ability to work under time pressure

#### Automatic Transmission Design — Personal Project

June 2026 - Present

- Designed a 4-speed automatic transmission from drivetrain theory to part design for automotive applications.
- Worked out planetary gear train dimensions, tooth counts and gear ratios, requiring analytical problem-solving.
- Sized shafts and other mechanical components by analyzing the loads and forces they would need to handle.

### SKILLS

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**Languages:** French: Native    English: Fluent - TOEIC: 925    German: Elementary    Spanish: Elementary

**Computer Skills:** CATIA V6, NX, Abaqus, Ansys Fluent, Ansys Granta EduPack, Adams, Python, C, Office Package, 3DEXperience, Sketchup 2022, HTML/CSS/JS

### INVOLVEMENT

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**Student Bar – Supmeca (President):** Managed stock and budget of €5000 and negotiated with commercial partners.

**Supmeca Humanitarian Association (Member):** Fundraising for underprivileged education access.

### MAIN INTERESTS

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**Music:** Piano (10 years in music school), Guitar (Self-Taught), Computed-assisted Music (FL Studio)

**Sport:** Football, Volleyball